

YEAR 7 MATHEMATICS · CHAPTER 6

Geometrical figures

6.01 Transforming shapes

6.03 Line symmetry

6.05 Classifying triangles

6.07 Exterior angle of a triangle

6.09 Angle sum of a quadrilateral

6.02 Combined transformations

6.04 Rotational symmetry

6.06 Angle sum of a triangle

6.08 Quadrilaterals

6.10 Properties of quadrilaterals

Name

Class

Teacher

How to use this booklet

- Each lesson starts with **learning intentions** and **success criteria**. Write the **key terms** as your teacher explains them, and use the **notes** box.
- Work through the levels in order: Level 1, then Level 2, then Level 3. Extension questions stretch you further.
- Some questions ask you to **draw** a diagram, graph or construction. Use the space given, a ruler and a sharp pencil.
- At the end of each lesson, **show off your skill**: choose one question, then tear off the ticket and hand it in.
- Missed a lesson? Each lesson ends with a **summary** that has a worked example for every level. Start there, then try the questions.

WE DO

Grey boxes with a dark border. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box and write your answer.

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In this chapter you will

- reflect, translate and rotate shapes
- find line and rotational symmetry
- classify triangles and quadrilaterals
- use the angle sums of triangles and quadrilaterals
- use the exterior angle of a triangle
- describe the properties of special quadrilaterals

NSW Mathematics K-10 Syllabus (2022), Stage 4 Measurement and space: MA4-ANG-C-01 (angle relationships, triangles and quadrilaterals) and transformations of shapes on the Cartesian plane and grids.



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
6.01

Transforming shapes

Date _____

Learning intentions

We are learning to:

- reflect, translate and rotate a shape on a grid

Success criteria

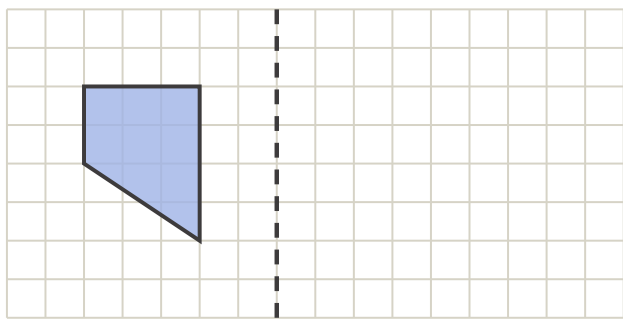
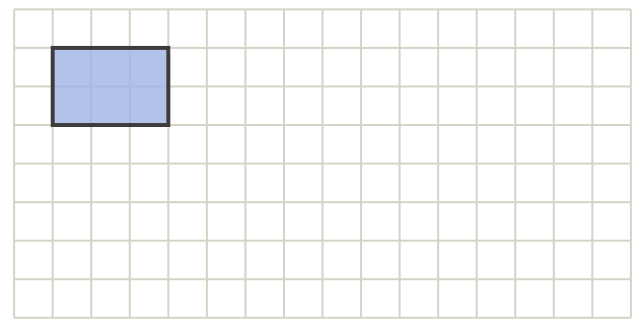
I can:

- ☐ **reflect** a shape across a mirror line
- ☐ **translate** a shape a given distance left, right, up or down
- ☐ **rotate** a shape 90° , 180° or 270° about a point

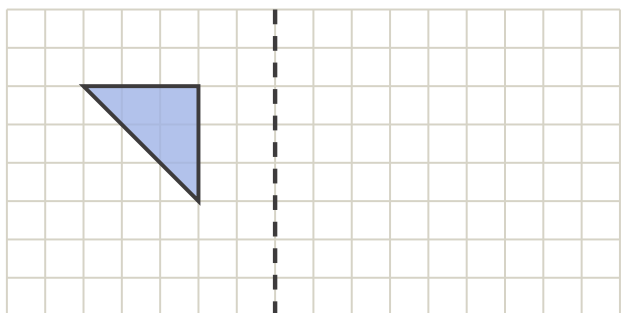
Key terms Write the definition as your teacher explains it.**Reflect****Translate****Rotate**

Notes

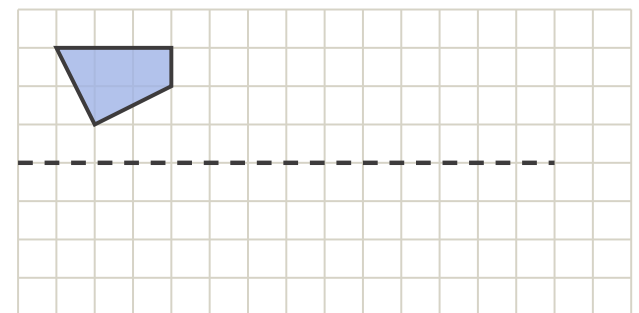
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Reflect the shape in the dotted line.**Example 2** Translate the shape 5 right and 3 down.**YOU DO** On your own: Set A. Reflect each shape in the dotted line.

1

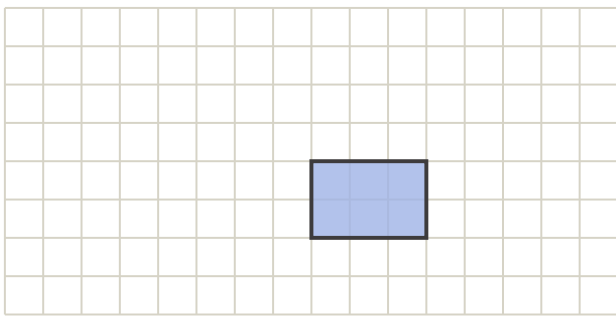


2



YOU DO On your own: Set B. Translate each shape.

1 4 units left, 1 unit up



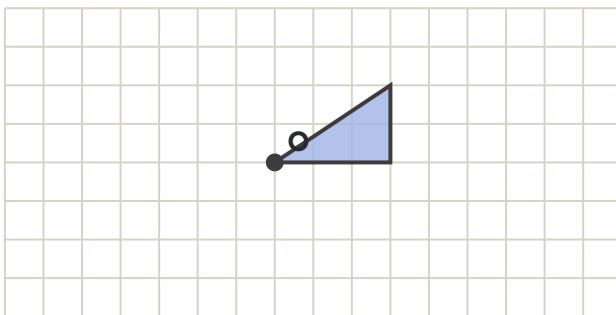
2 4 units right, 6 units up



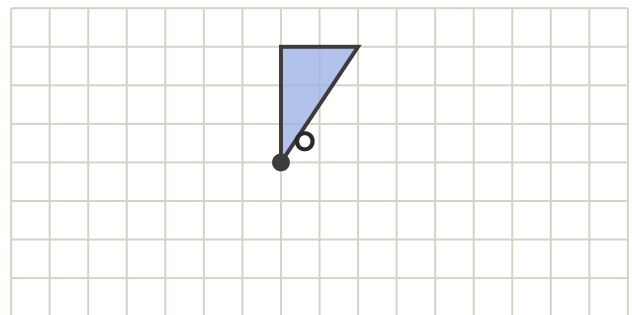
Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Rotate 180° about O.

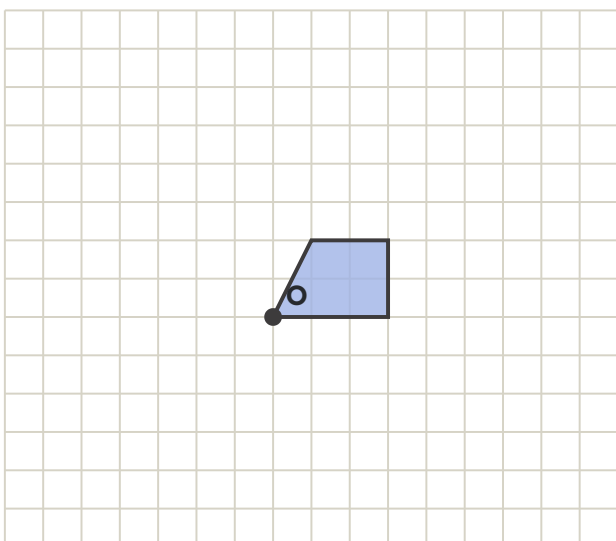


Example 4 Rotate 90° clockwise about O.

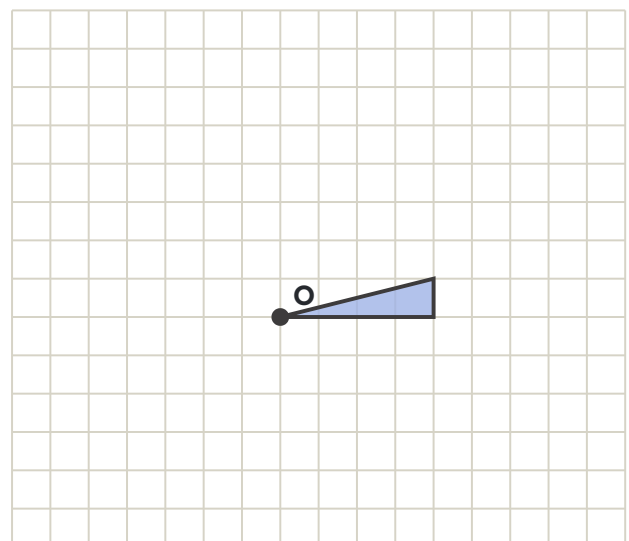


YOU DO On your own: Set C. Rotate each shape about O.

1 180°



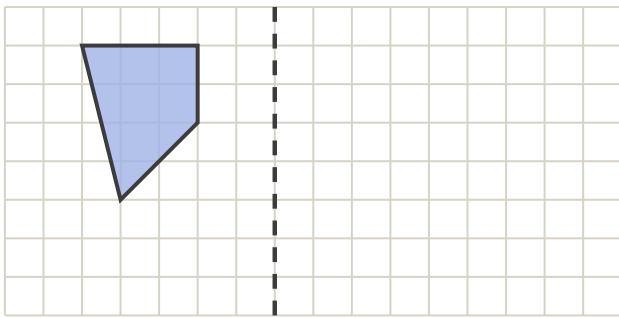
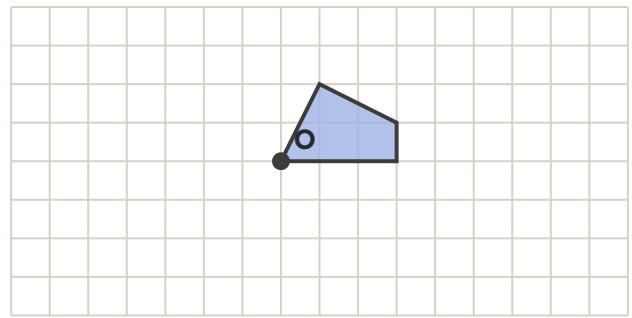
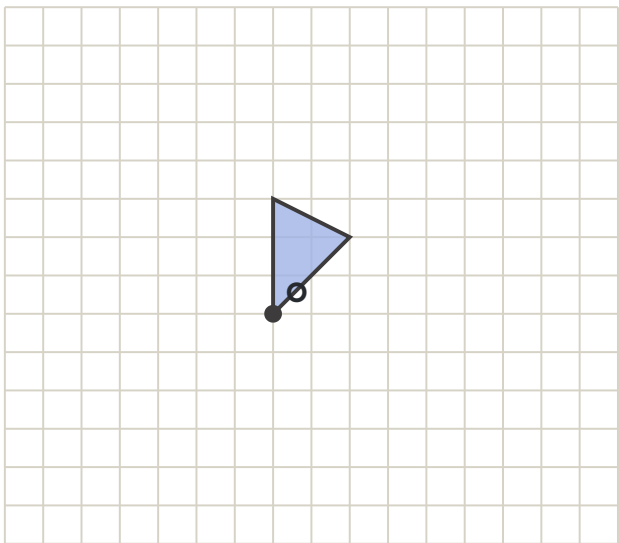
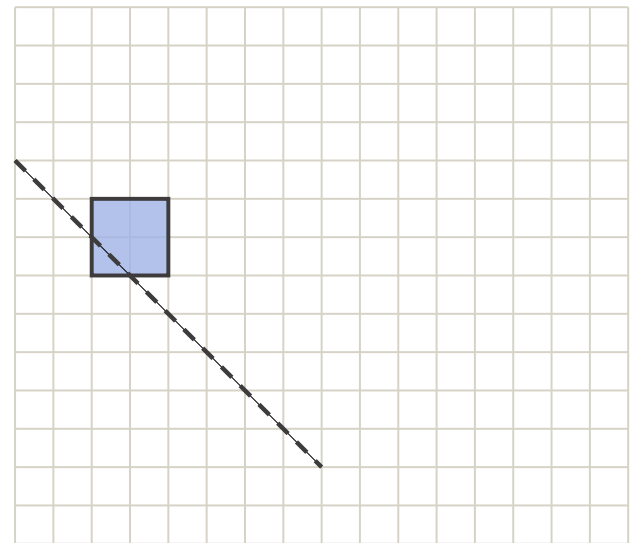
2 90° clockwise





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Reflect in the dotted line.**Example 6** Rotate 270° clockwise about O.**YOU DO** On your own: Set D. Transform each shape.**1** Rotate 270° clockwise about O.**2** Reflect in the dotted line.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Name the transformation that flips a shape.

Level 2

A point is at (2, 3) on a grid. Where is it after a translation 4 right and 1 down?

Level 3Describe what a 90° clockwise rotation does to an arrow pointing up.

Extension · Stretch yourself

Worked example

A shape is rotated 90° clockwise and then 180° . What single rotation does the same thing?

1. Rotations about the same point add
2. $90^\circ + 180^\circ = 270^\circ$

Answer: 270° clockwise (or 90° anticlockwise)

YOU DO

On your own: read the worked example, then try these.

- E1** Which single transformation is the same as reflecting in a vertical line and then in a horizontal line that crosses it?

- E2** Is a 90° clockwise rotation the same as a 270° anticlockwise rotation? Explain.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. **Reflect:** flip over the mirror line; same distance each side.
 2. **Translate:** slide; count units left/right then up/down.
 3. **Rotate:** turn about a point; use tracing paper.
 4. The shape keeps its size and shape.

Level 1

Reflect in a vertical line

1. Each vertex the same distance on the other side

Answer: a mirror image

Level 2

Translate 5 right, 3 down

1. Move every vertex

Answer: same shape, new position

Level 3

Rotate 180° about O

1. Each vertex goes through O to the other side

Answer: upside down

Extension

90° then 180°

1. Add

Answer: 270° clockwise

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.02

Combined transformations

Date _____

Learning intentions

We are learning to:

- apply two or more transformations in order
- describe a combined transformation

Success criteria

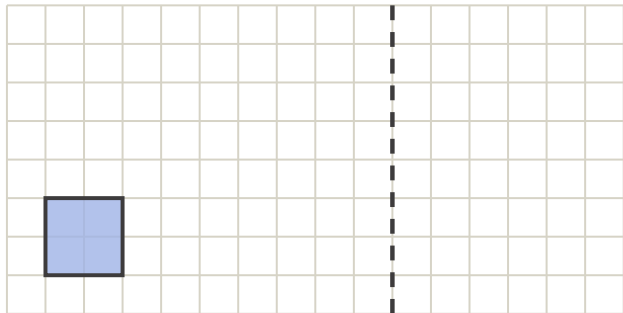
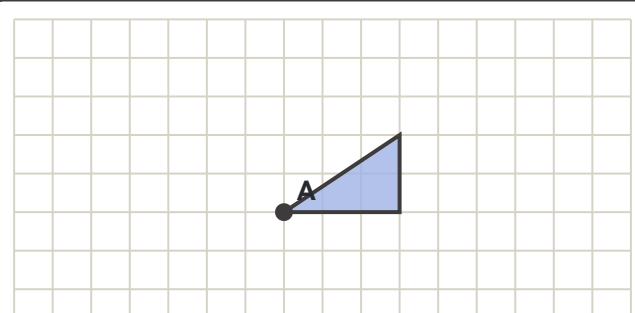
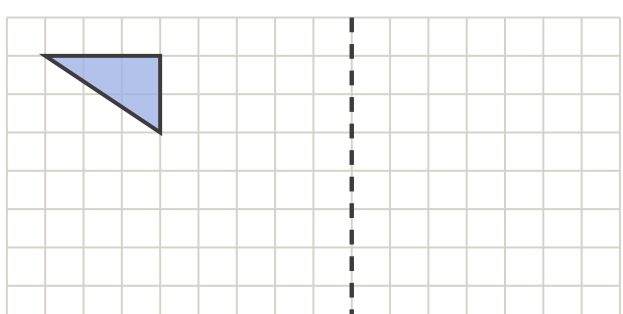
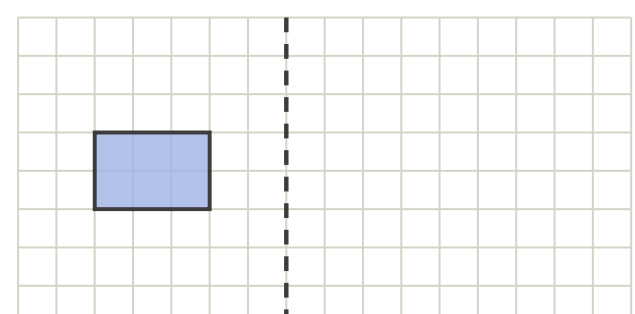
I can:

- ☐ do each transformation in the order given
- ☐ label each new image (A , A' , A'')
- ☐ describe a combination that maps one shape onto another

Key terms Write the definition as your teacher explains it.**Image****Combined transformation**

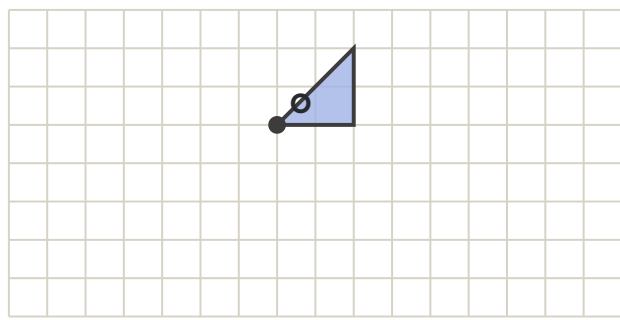
Notes

Level 1 · Getting started

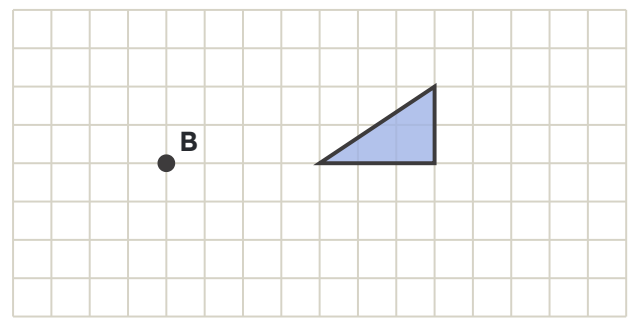
WE DO With your teacher: copy what they write on the board.**Example 1** Translate 5 right, 2 up; then reflect.**Example 2** Rotate 90° clockwise about A, then 3 left.**YOU DO** On your own: Set A. Do both transformations.**1** Translate 3 right; then reflect.**2** Reflect; then translate 2 up.

YOU DO On your own: Set B. Do both transformations.

- 1 Rotate 180° about O, then translate 2 down.



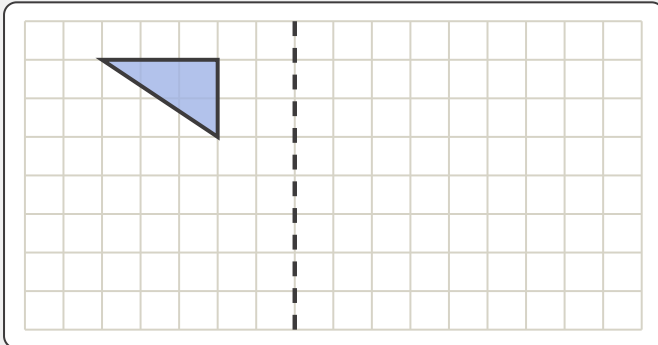
- 2 Translate 4 left, then rotate 90° clockwise about B.



Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

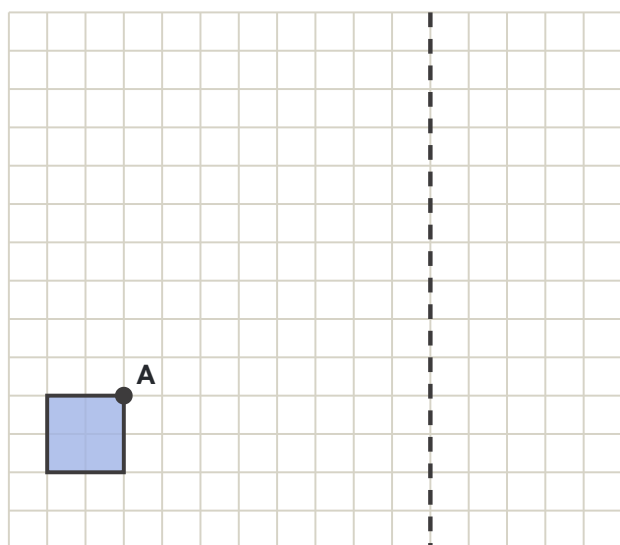
- Example 3** Reflect in the dotted line, then translate 1 down.



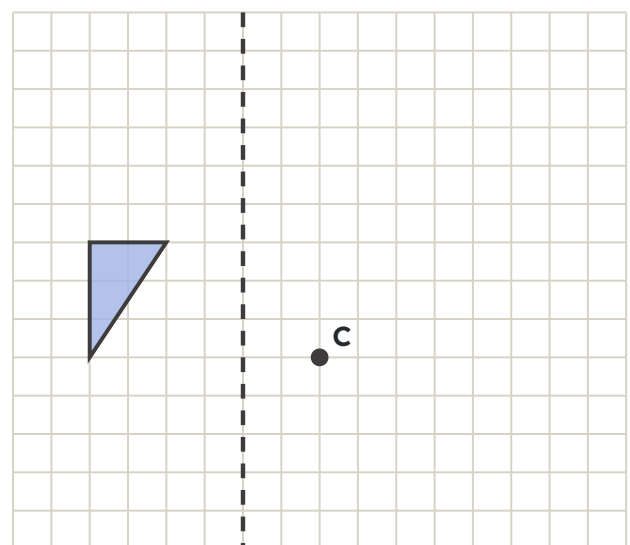
- Example 4** Does the order matter? Try reflecting then translating versus translating then reflecting.

YOU DO On your own: Set C. Do all three transformations.

- 1 Translate 5 right, 2 up; rotate 90° clockwise about A; reflect in the dotted line.



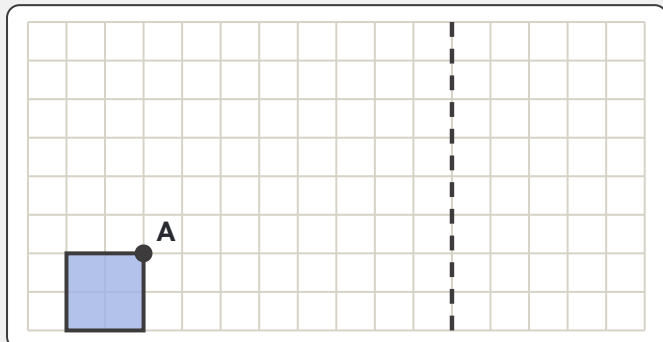
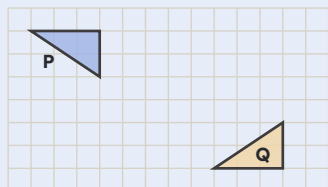
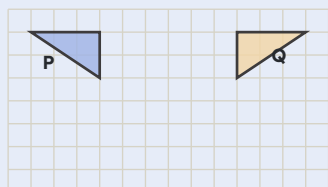
- 2 Reflect in the dotted line; translate 1 left; rotate 180° about C.





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Translate 5 right and 2 up, rotate 90° clockwise about A, then reflect in the dotted line.**Example 6** Describe two transformations that move a shape from the top-left to the bottom-right of a grid, upside down.
YOU DO On your own: Set D. Describe the transformations.**1** Describe how shape P moves onto shape Q.**3** Does a translation followed by a reflection give the same result as the reflection first? Give an example.
2 Find two different ways to move P onto Q.**4** Which transformations never change the direction a shape faces?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What does A' mean?

Level 2

A point (1, 2) is translated 3 right then 4 up. Where is it?

Level 3

A point (2, 3) is reflected in the x-axis then translated 2 left. Where is it?

Extension · Stretch yourself

Worked example

A triangle has vertices $(1, 1)$, $(3, 1)$, $(3, 4)$. It is reflected in the y -axis, then translated 2 units down. Where are the vertices?

1. Reflect in y -axis: $(-1, 1)$, $(-3, 1)$, $(-3, 4)$
2. Down 2: subtract 2 from each y

Answer: $(-1, -1)$, $(-3, -1)$, $(-3, 2)$

YOU DO

On your own: read the worked example, then try these.

- E1** Reflect $(2, 5)$ in the x -axis, then translate 3 left. Where is it?

- E2** Rotate $(4, 1)$ by 180° about the origin. Where is it?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Do the transformations in the order given.
 3. Label them A , A' , A'' .

2. Draw each image before doing the next.
4. Order can change the final position.

Level 1

Translate then reflect

1. Slide first
2. Then flip over the line

Answer: image A''

Level 2

Rotate then translate

1. Turn about the point
2. Then slide

Answer: image A''

Level 3

Order matters?

1. Try both ways

Answer: often yes

Extension

Reflect in y -axis then down 2

1. $(x, y) \rightarrow (-x, y - 2)$

Answer: $(-1, -1)$, $(-3, -1)$, $(-3, 2)$

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LESSON
6.03

Line symmetry

Date _____

Learning intentions

We are learning to:

- find the axes of symmetry of a shape

Success criteria

I can:

- ☐ know that an axis of symmetry divides a shape into two mirror images
- ☐ draw every axis of symmetry on a shape
- ☐ count the axes of symmetry

Key terms Write the definition as your teacher explains it.**Line symmetry****Axis of symmetry**

Notes

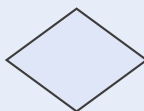
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Draw the axes of symmetry.**Example 2** Draw the axes of symmetry.**YOU DO** On your own: Set A. Draw all the axes of symmetry. Write how many.

1



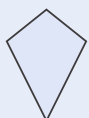
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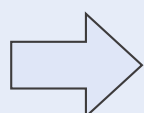
5



2



4

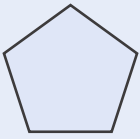


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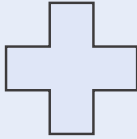


YOU DO On your own: Set B. Draw all the axes of symmetry. Write how many.

1



2



3



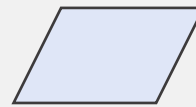
Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 How many axes?



Example 4 How many axes?



YOU DO On your own: Set C. Draw all the axes of symmetry. Write how many.

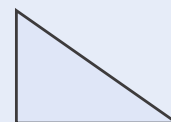
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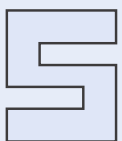
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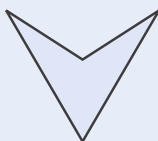
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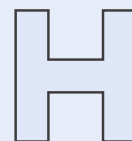
2



4



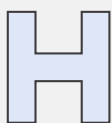
6





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** How many axes?**Example 6** How many axes of symmetry does a circle have?

YOU DO On your own: Set D. Think it through.

- 1 How many axes of symmetry does a regular 10-sided shape have?

- 3 Draw a shape with exactly 2 axes of symmetry.

- 2 Which capital letters of the alphabet have exactly one axis of symmetry? List 5.

- 4 Can a triangle have exactly 2 axes of symmetry? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

How many axes of symmetry does an equilateral triangle have?

Level 2

How many axes of symmetry does a kite have?

Level 3

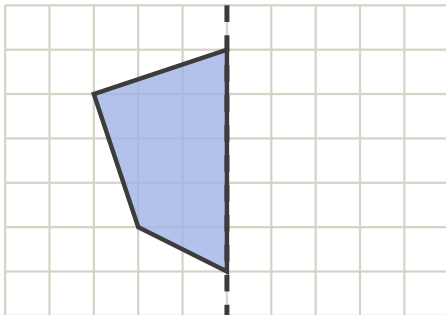
How many axes of symmetry does a regular hexagon have?

Extension · Stretch yourself

Worked example

Complete the shape so that the dotted line is an axis of symmetry.

- Each point is reflected across the line
- Join the reflected points

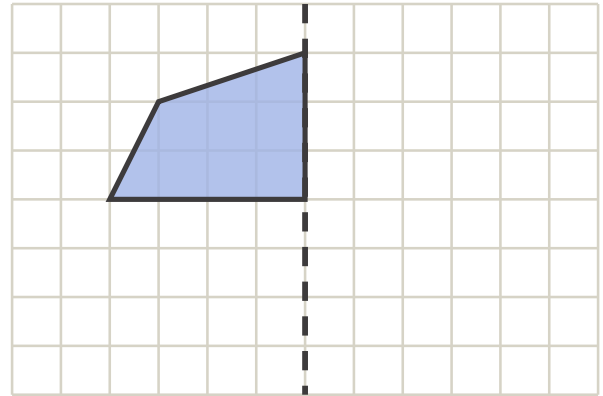


Answer: a symmetric shape

YOU DO

On your own: read the worked example, then try these.

- E1** Complete the shape so both dotted lines are axes of symmetry.



- E2** A regular polygon has 12 axes of symmetry. How many sides does it have?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. An **axis of symmetry** folds a shape onto itself.
 2. Test by folding or with a mirror.
 3. Draw the axes as dotted lines.
 4. A regular polygon with n sides has n axes.

Level 1

Rectangle

1. Vertical and horizontal only

Answer: 2 axes

Level 2

Square

1. 2 through sides, 2 diagonals

Answer: 4 axes

Level 3

Parallelogram

1. No fold works

Answer: 0 axes

Extension

Regular 12-gon

1. n axes

Answer: 12 axes

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.04

Rotational symmetry

Date _____

Learning intentions

We are learning to:

- find the order of rotational symmetry of a shape

Success criteria

I can:

- ☐ know a shape has rotational symmetry if it looks the same before a full turn
- ☐ count the **order**: how many times it matches in one full turn
- ☐ find the angle of each turn

Key terms Write the definition as your teacher explains it.

Rotational symmetry

Order

Notes

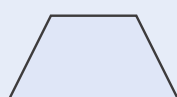
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Order of rotational symmetry?**Example 2** Order?**YOU DO** On your own: Set A. Does the shape have rotational symmetry? If so, give its order.

1



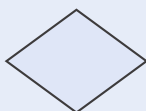
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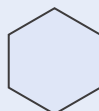
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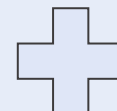
2



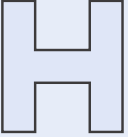
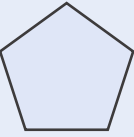
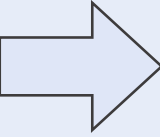
4



6



YOU DO On your own: Set B. Give the order of rotational symmetry.

1 	2 	3 
		

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

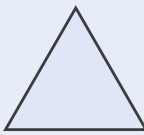
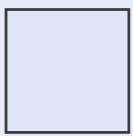
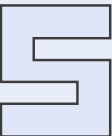
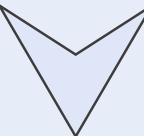
Example 3 Order?



Example 4 Order?



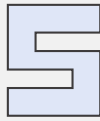
YOU DO On your own: Set C. Give the order and the smallest angle of turn.

1 	3 	5 
		
2 	4 	6 
		



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Order?**Example 6** A shape has order 5. What is the smallest angle it turns to match itself?

YOU DO On your own: Set D. Think it through.

- 1 Which capital letters have rotational symmetry of order 2? List 4.

- 3 A shape matches itself every 72° . What is its order?

- 2 Can a shape have rotational symmetry but no line symmetry? Give an example.

- 4 Draw a shape with rotational symmetry of order 4.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is the order of rotational symmetry of a square?

Level 2

Does a kite have rotational symmetry?

Level 3

A shape has order 6. Find its smallest angle of turn.

Extension · Stretch yourself

Worked example

A regular polygon matches itself every 40° . How many sides does it have?

1. A full turn is 360°
2. $360 \div 40 = 9$

Answer: 9 sides

YOU DO

On your own: read the worked example, then try these.

- E1** A regular polygon matches itself every 30° . How many sides does it have?

- E2** Which has the larger angle of turn: order 6 or order 8? Explain.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Turn the shape about its centre.
 2. Count how many times it looks the same in one turn.
 3. That count is the **order**.
 4. Order 1 means no rotational symmetry. Angle = $360^\circ \div \text{order}$.

Level 1

Rectangle

1. Matches at 180° and 360°

Answer: order 2

Level 2

Square

1. Every 90°

Answer: order 4

Level 3

Kite

1. Only at 360°

Answer: no rotational symmetry

Extension

Every 40°

1. $360 \div 40$

Answer: 9 sides

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.05

Classifying triangles

Date _____

Learning intentions

We are learning to:

- classify triangles by their sides and by their angles

Success criteria

I can:

- ☐ name triangles by sides: equilateral, isosceles, scalene
- ☐ name triangles by angles: acute-angled, right-angled, obtuse-angled
- ☐ find equal angles in an isosceles triangle

Key terms Write the definition as your teacher explains it.

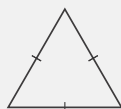
Isosceles

Scalene

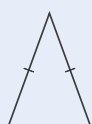
Equilateral

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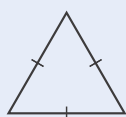
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Classify by sides and by angles.**Example 2** Classify by sides and by angles.**YOU DO** On your own: Set A. Classify each triangle by its sides.

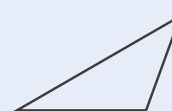
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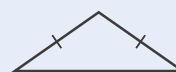
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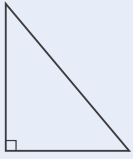


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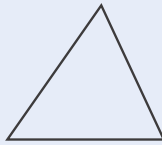


YOU DO On your own: Set B. Classify each triangle by its angles.

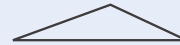
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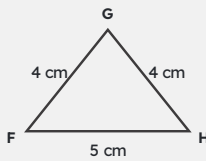
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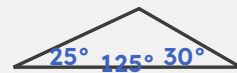
Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Classify $\triangle FGH$. Which angles are equal?

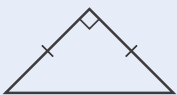


Example 4 Classify by sides and angles.

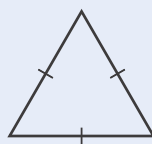


YOU DO On your own: Set C. Classify each triangle by sides and by angles.

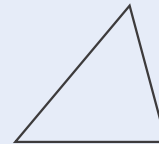
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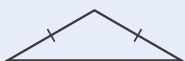
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Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** My angles are 60° , 80° and 40° .
Classify me.

Example 6 All my angles are equal. Classify me.

YOU DO On your own: Set D. Classify each described triangle by sides and by angles.**1** My angles are 60° , 80° and 40° .

3 My angles are 30° , 30° and 120° .

2 My sides are 5 cm, 5 cm and 7 cm, and one angle is 90° . Is this possible? Explain.

4 Can a triangle be right-angled and equilateral? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Classify a triangle with sides 6, 6 and 6 cm.

Level 2Classify a triangle with angles 90° , 45° and 45° .

Level 3Classify a triangle with angles 110° , 35° and 35° by sides and angles.

Extension · Stretch yourself

Worked example

An isosceles triangle has one angle of 40° . What could the other two angles be?

1. If 40° is one of the equal angles: $40^\circ, 40^\circ, 100^\circ$
2. If 40° is the apex: $(180 - 40) \div 2 = 70^\circ$

Answer: 40° and 100° , or 70° and 70°

YOU DO

On your own: read the worked example, then try these.

- E1** An isosceles triangle has one angle of 100° . Find the other two angles.

- E2** An isosceles triangle has one angle of 60° . What type of triangle is it?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Sides: **equilateral** (3 equal), **isosceles** (2 equal), **scalene** (none).
 2. Angles: **acute** (all $< 90^\circ$), **right** (one 90°), **obtuse** (one $> 90^\circ$).
 3. Equal sides have matching tick marks.
 4. Angles opposite equal sides are equal.

Level 1

3 equal sides

1. All angles 60°

Answer: equilateral, acute

Level 2

$\triangle FGH$: 4 cm, 4 cm, 5 cm

1. Two equal sides

Answer: isosceles; $\angle F = \angle H$

Level 3

$25^\circ, 30^\circ, 125^\circ$

1. One obtuse angle

Answer: scalene, obtuse

Extension

Isosceles with a 40° angle

1. Two cases

Answer: $40^\circ, 100^\circ$ or $70^\circ, 70^\circ$

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LESSON
6.06

Angle sum of a triangle

Date _____

Learning intentions

We are learning to:

- use the angle sum of a triangle to find unknown angles

Success criteria

I can:

- ☐ know the angles in a triangle add to 180°
- ☐ use equal angles in isosceles and equilateral triangles
- ☐ give reasons for each step

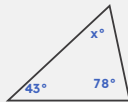
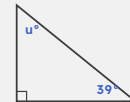
Key terms Write the definition as your teacher explains it.

Angle sum

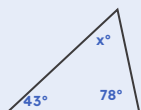
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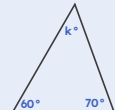
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find x .**Example 2** Find u .**YOU DO** On your own: Set A. Find the value of each variable.

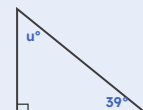
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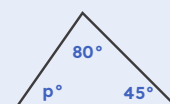
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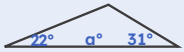


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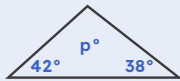


YOU DO On your own: Set B. Find the value of each variable.

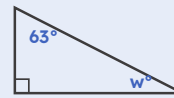
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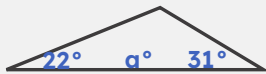




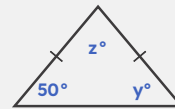
Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Find a .

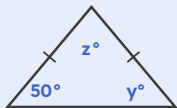


Example 4 Find y and z .

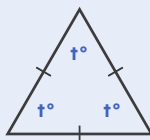


YOU DO On your own: Set C. Isosceles and equilateral triangles. Find each variable.

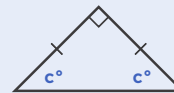
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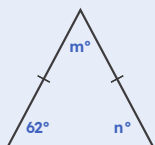
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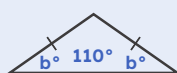
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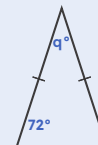
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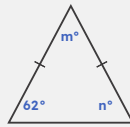
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Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find n and m (isosceles).**Example 6** Find $\angle PQR$.**YOU DO** On your own: Set D. Find the unknowns. Give reasons.

- 1 The angles of a triangle are x° , $2x^\circ$ and $3x^\circ$. Find x .

- 3 Can a triangle have angles of 100° and 90° ? Explain.

- 2 An isosceles triangle has an apex angle of 36° . Find the base angles.

- 4 Two angles of a triangle are equal and the third is 20° more than each of them. Find all three.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Two angles of a triangle are 65° and 45° . Find the third.
Level 2A right-angled triangle has an angle of 28° . Find the other angle.
Level 3An isosceles triangle has base angles of 67° . Find the apex angle.

Extension · Stretch yourself

Worked example

Prove that the angles of a triangle add to 180° using a line parallel to the base.

1. Draw a line through the top vertex parallel to the base
2. The two outer angles equal the base angles (alternate angles)
3. The three angles at the top form a straight line: 180°

Answer: $a + b + c = 180^\circ$

YOU DO

On your own: read the worked example, then try these.

- E1** Tear the corners off a paper triangle and place them together. What do you see?

- E2** Use the angle sum to explain why a triangle can have only one obtuse angle.

Lesson summary

Missed the lesson? Start here.

Remember 1. Angles in a triangle add to 180° .
3. Isosceles: base angles are equal.

2. Add the known angles, subtract from 180° .
4. Equilateral: every angle is 60° .

Level 1

$43^\circ, 78^\circ, x^\circ$

1. $180 - 43 - 78$

Answer: $x = 59$

Level 2

Right angle and 39°

1. $180 - 90 - 39$

Answer: $u = 51$

Level 3

Isosceles, apex 72° at P? no: base 72°

1. $\angle PQR = 180 - 72 - 72$

Answer: 36°

Extension

Proof

1. Parallel line and alternate angles

Answer: 180°

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.07

Exterior angle of a triangle

Date _____

Learning intentions

We are learning to:

- use the exterior angle property of a triangle

Success criteria

I can:

- ☐ identify an **exterior angle** of a triangle
- ☐ know the exterior angle equals the sum of the two interior opposite angles
- ☐ find unknown angles using this rule

Key terms Write the definition as your teacher explains it.**Exterior angle****Interior opposite angles**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find x .**Example 2** Find x .**YOU DO** On your own: Set A. Find the value of each variable.

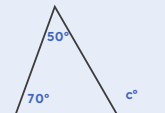
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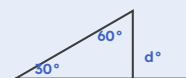
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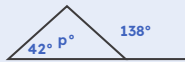


YOU DO On your own: Set B. Find the value of each variable.

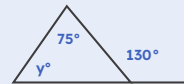
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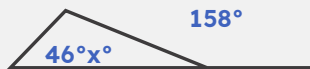




Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Find x .

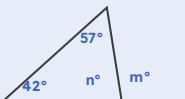


Example 4 Find p .

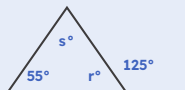


YOU DO On your own: Set C. Find each variable. Give reasons.

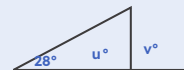
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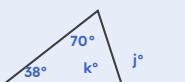
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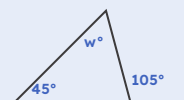
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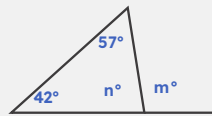
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Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find m and n .**Example 6** Why is the exterior angle always larger than each interior opposite angle?**YOU DO** On your own: Set D. Harder problems.

- 1 The interior opposite angles are x° and $2x^\circ$. The exterior angle is 111° . Find x .

- 3 What do the three exterior angles of a triangle (one at each vertex) add to? Test with an equilateral triangle.

- 2 The exterior angle of an isosceles triangle at the base is 110° . Find all three interior angles.

- 4 Can an exterior angle be acute? Explain.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Interior opposite angles 40° and 75° . Find the exterior angle.
Level 2The exterior angle is 125° and one interior opposite angle is 60° . Find the other.
Level 3The exterior angle is 140° . Find the adjacent interior angle.

Extension · Stretch yourself

Worked example

Prove the exterior angle property using the angle sum.

- $a + b + c = 180$ (angle sum)
- $c + x = 180$ (straight line)
- So $x = a + b$

Answer: The exterior angle equals the sum of the interior opposite angles

YOU DO

On your own: read the worked example, then try these.

- E1** An exterior angle is 4 times the interior angle next to it. Find the exterior angle.

- E2** The exterior angle is 130° and one interior opposite angle is 20° more than the other. Find both.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. An **exterior angle** is made by extending one side.
 2. Exterior angle = sum of the two **interior opposite** angles.
 3. Exterior + adjacent interior = 180° .
 4. Give the reason: "exterior angle of a triangle".

Level 1

59° and 56° opposite

- $59 + 56$

Answer: $x = 115$

Level 2

$$46^\circ + x = 158^\circ$$

- $158 - 46$

Answer: $x = 112$

Level 3

$$42^\circ + p = 138^\circ$$

- $138 - 42$

Answer: $p = 96$

Extension

Proof

1. Angle sum and straight line

Answer: $x = a + b$

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LESSON
6.08

Quadrilaterals

Date _____

Learning intentions

We are learning to:

- classify quadrilaterals
- describe the symmetry and definitions of special quadrilaterals

Success criteria

I can:

- ☐ decide if a quadrilateral is **convex** or **non-convex**
- ☐ name the special quadrilaterals
- ☐ give the definition of each special quadrilateral

Key terms Write the definition as your teacher explains it.**Convex****Parallelogram****Rhombus**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Convex or non-convex?**Example 2** Name this quadrilateral.**YOU DO** On your own: Set A. Convex or non-convex?

1



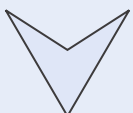
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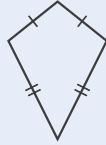


YOU DO On your own: Set B. Name each special quadrilateral.

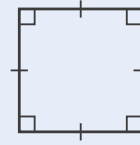
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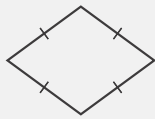
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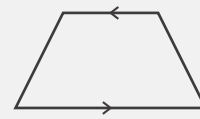
Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Name this quadrilateral.



Example 4 Name this quadrilateral.



YOU DO On your own: Set C. For each shape in Set B and Examples 2–4, write the number of axes of symmetry and the order of rotational symmetry.

1 Rectangle

3 Square

5 Rhombus

2 Kite

4 Parallelogram

6 Trapezium (isosceles)



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** What is the definition of a rhombus?
Example 6 What am I? I have one pair of parallel sides.
YOU DO On your own: Set D. Definitions.**1** What quadrilateral am I? I have both pairs of opposite sides parallel.
3 Is a square a rhombus? Explain using definitions.
2 Write the definition of a rectangle.
4 Is every rectangle a parallelogram? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Is a square convex?

Level 2

Name a quadrilateral with exactly one pair of parallel sides.

Level 3

What is the definition of a parallelogram?

Extension · Stretch yourself

Worked example

Draw a family tree of the quadrilaterals.

1. Quadrilateral → trapezium → parallelogram
2. Parallelogram → rectangle and rhombus
3. Rectangle and rhombus → square

Answer: A square is a special rectangle and a special rhombus

YOU DO

On your own: read the worked example, then try these.

- E1** Where does a kite fit in the family tree? Is a rhombus a kite?

- E2** Which special quadrilaterals have exactly 2 axes of symmetry?

Lesson summary

Missed the lesson? Start here.

- Definitions**
1. **Trapezium:** at least one pair of parallel sides.
 2. **Rhombus:** parallelogram with all sides equal.
 3. **Parallelogram:** both pairs parallel.
 - Rectangle:** parallelogram with a right angle.
 3. **Square:** rectangle with all sides equal. **Kite:** two pairs of equal adjacent sides.
 4. **Convex:** all angles less than 180° .

Level 1

Dart shape

1. One angle is reflex

Answer: non-convex

Level 2

Opposite sides parallel

1. Both pairs

Answer: parallelogram

Level 3

Rhombus definition

1. Parallelogram, all sides equal

Answer: a parallelogram with 4 equal sides

Extension

Square in the family

1. Rectangle and rhombus

Answer: both

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
6.09

Angle sum of a quadrilateral

Date _____

Learning intentions

We are learning to:

- use the angle sum of a quadrilateral to find unknown angles

Success criteria

I can:

- ☐ know the angles in a quadrilateral add to 360°
- ☐ find an unknown angle
- ☐ use right angles and equal angles in special quadrilaterals

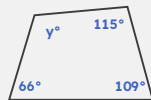
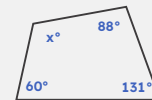
Key terms Write the definition as your teacher explains it.

Angle sum

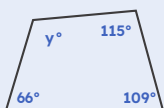
Quadrilateral

Notes

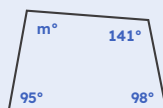
Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Find y .**Example 2** Find x .**YOU DO** On your own: Set A. Find the value of each variable.

1



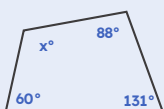
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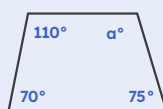
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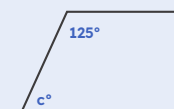
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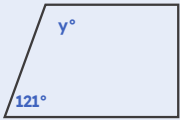


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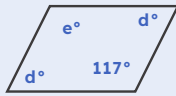


YOU DO On your own: Set B. Find each variable.

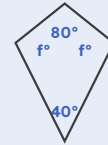
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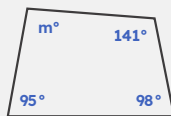
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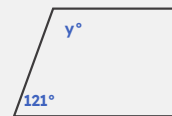
Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Find m .

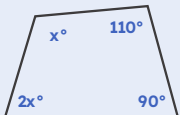


Example 4 Find y .

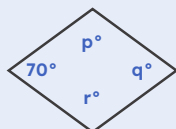


YOU DO On your own: Set C. Find each variable. Give reasons.

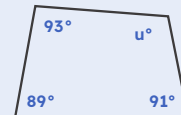
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3



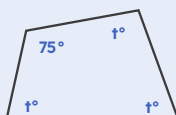
5



2



4



6





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Why do the angles in a quadrilateral add to 360° ?
Example 6 Three angles of a quadrilateral are equal and the fourth is 90° . Find the equal angles.
YOU DO On your own: Set D. Think it through.**1** The angles of a quadrilateral are x° , $2x^\circ$, $3x^\circ$ and $4x^\circ$. Find each angle.
3 A parallelogram has one angle of 58° . Find the other three.
2 Can a quadrilateral have 4 obtuse angles? Explain.
4 A kite has angles 120° and 60° between the unequal sides. Find the other two (equal) angles.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1Three angles are 90° , 90° and 70° . Find the fourth.
Level 2Three angles are 100° , 85° and 95° . Find the fourth.
Level 3A parallelogram has an angle of 65° . Find the other angles.

 Extension · Stretch yourself
 Worked example

Find the angle sum of a pentagon by splitting it into triangles.

1. Draw diagonals from one vertex: 3 triangles
2. $3 \times 180^\circ$

Answer: 540°

YOU DO

On your own: read the worked example, then try these.

- E1** Find the angle sum of a hexagon.

- E2** Find each angle of a regular pentagon.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Angles in a quadrilateral add to 360° .
 3. Add the known angles, subtract from 360° .

2. A diagonal makes 2 triangles: $2 \times 180^\circ$.
4. Use properties: opposite angles in a parallelogram are equal.

 Level 1

$66^\circ, 109^\circ, 115^\circ, y^\circ$

1. $360 - 290$

Answer: $y = 70$

 Level 2

$60^\circ, 131^\circ, 88^\circ, x^\circ$

1. $360 - 279$

Answer: $x = 81$

 Level 3

Two right angles and 121°

1. $360 - 90 - 90 - 121$

Answer: $y = 59$

 Extension

Pentagon

1. 3 triangles

Answer: 540°

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LESSON
6.10

Properties of quadrilaterals

Date _____

Learning intentions

We are learning to:

- describe the side, angle and diagonal properties of special quadrilaterals

Success criteria

I can:

- ☐ list the properties of a parallelogram
- ☐ describe how the diagonals behave in each special quadrilateral
- ☐ identify a quadrilateral from its properties

Key terms Write the definition as your teacher explains it.**Diagonal****Bisect**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** List 2 properties of a parallelogram.
Example 2 EFGH is a parallelogram. Which is NOT true? A $\angle EFG = \angle EHG$ B $EF = HG$ C $EF = FG$ D $\angle EFG + \angle FGH = 180^\circ$

YOU DO On your own: Set A. True or false?

1 A rectangle has equal diagonals.

2 A rhombus has 4 right angles.

3 The diagonals of a square are perpendicular.

4 A kite has 2 pairs of parallel sides.

5 A parallelogram has opposite sides equal.

6 A trapezium always has equal diagonals.

YOU DO On your own: Set B. Name all the quadrilaterals that have:

1 4 equal sides

2 4 right angles

3 opposite angles equal

4 diagonals perpendicular

5 exactly one pair of parallel sides

6 diagonals that bisect the angles

Level 2 · Building up

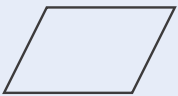
WE DO With your teacher: copy what they write on the board.

Example 3 What am I? All my angles are equal.

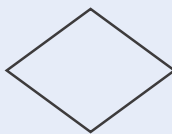
Example 4 What am I? My diagonals are equal and bisect each other.

YOU DO On your own: Set C. Draw the diagonals and describe them (equal? bisect? perpendicular?).

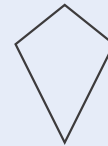
1 Parallelogram



3 Rhombus



5 Kite



2 Rectangle



4 Square



6 Trapezium





Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** What am I? My diagonals bisect each other.
Example 6 What am I? My opposite angles are equal.
YOU DO On your own: Set D. Use properties.**1** ABCD is a parallelogram with $AB = 7$ cm, $BC = 4$ cm and $\angle A = 65^\circ$. Find CD, AD and $\angle C$.
3 What am I? My diagonals are equal, bisect each other and are perpendicular.
2 The diagonals of a rhombus are 6 cm and 8 cm. What angle do they meet at?
4 What am I? I have one axis of symmetry and diagonals that are perpendicular.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Name a quadrilateral with equal diagonals.

Level 2

State one property of the diagonals of a rhombus.

Level 3What am I? My opposite sides are equal and all angles are 90° .

Extension · Stretch yourself

Worked example

A rhombus has diagonals 10 cm and 24 cm. Find its side length.

1. Diagonals bisect each other at right angles: halves are 5 and 12
2. $\text{Side}^2 = 5^2 + 12^2 = 169$ (Pythagoras, from Stage 5)

Answer: 13 cm

YOU DO

On your own: read the worked example, then try these.

- E1** A rectangle has sides 6 cm and 8 cm. Its diagonals are 10 cm. How far is the centre from each corner?

- E2** Explain why a square's diagonals make angles of 45° with its sides.

Lesson summary

Missed the lesson? Start here.

- Diagonals**
1. Parallelogram: diagonals **bisect** each other.
 2. Rectangle: diagonals equal and bisect.
 3. Rhombus: diagonals bisect at 90° and bisect the angles.
 4. Square: all of these. Kite: one diagonal bisects the other at 90° .

Level 1

Parallelogram properties

1. Opposite sides equal
2. Opposite angles equal

Answer: any 2

Level 2

EFGH: which is not true?

1. Adjacent sides need not be equal

Answer: C

Level 3

Diagonals equal and bisect

1. Rectangle family

Answer: rectangle, square

Extension

Rhombus diagonals 10, 24

1. Halves 5, 12

Answer: 13 cm

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